
YABAI

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YABAI (Yet Another Bühlmann Algorithm Implementation) is a Python package to plan scuba dives using the Bühlmann ZH-L16 (A, B or C) models

DISCLAIMER

This project is intended only for academic purposes.

DO NOT USE IT FOR REAL DIVES PLANNING

None of the authors, contributors, administrators, or anyone else connected with this project can be responsible for your use of the information provided by this application or linked from these web pages. Use the results at your own risk.

CHAPTER
TWO

USAGE

Lorem ipsum

DECOMPRESSION THEORY

Lorem ipsum

IMPLEMENTATION DETAILS

Lorem ipsum

API REFERENCE

<i>yabai</i>	YABAI - Yet Another Bühlmann Algorithm Implementation
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5.1 yabai

YABAI - Yet Another Bühlmann Algorithm Implementation

Modules

<i>constants</i>	General constants definitions
<i>exceptions</i>	Custom exceptions
<i>profile</i>	Dive profile definitions
<i>tanks</i>	Tank and gas definitions.

5.1.1 yabai.constants

General constants definitions

Module Attributes

<i>ZH_L16</i>	Bühlmann ZH-L16 coefficients.
<i>NOAA_CNS_POINTS</i>	NOAA curve points for CNS calculations.
<i>NOAA_CNS_EQUATIONS</i>	Linealized form of the NOAA CNS curve.
<i>PW</i>	Water vapour pressure.

5.1.1.1 yabai.constants.ZH_L16

```
yabai.constants.ZH_L16 = {'C': {'He': {'a': array([1.6189, 1.383 , 1.1919, 1.0458,
0.922 , 0.8205, 0.7305, 0.6502, 0.595 , 0.5545, 0.5333, 0.5189, 0.5181, 0.5176, 0.5172,
0.5119]), 'b': array([0.477 , 0.5747, 0.6527, 0.7223, 0.7582, 0.7957, 0.8279, 0.8553,
0.8757, 0.8903, 0.8997, 0.9073, 0.9122, 0.9171, 0.9217, 0.9267]), 'ht': array([ 1.88,
3.02, 4.72, 6.99, 10.21, 14.48, 20.53, 29.11, 41.2 , 55.19, 70.69, 90.34, 115.29, 147.42,
188.24, 240.03])}, 'N2': {'a': array([1.1696, 1. , 0.8618, 0.7562, 0.62 , 0.5043, 0.441
, 0.4 , 0.375 , 0.35 , 0.3295, 0.3065, 0.2835, 0.261 , 0.248 , 0.2327]), 'b':
array([0.5578, 0.6514, 0.7222, 0.7825, 0.8126, 0.8434, 0.8693, 0.891 , 0.9092, 0.9222,
0.9319, 0.9403, 0.9477, 0.9544, 0.9602, 0.9653]), 'ht': array([ 5. , 8. , 12.5, 18.5,
27. , 38.3, 54.3, 77. , 109. , 146. , 187. , 239. , 305. , 390. , 498. , 635. ])}}}
```

Bühlmann ZH-L16 coefficients.

5.1.1.2 `yabai.constants.NOAA_CNS_POINTS`

```
yabai.constants.NOAA_CNS_POINTS = {0.5: 900.0, 0.6: 720.0, 0.7: 570.0, 0.8: 450.0,  
0.9: 360.0, 1.0: 300.0, 1.1: 270, 1.2: 240, 1.3: 210.0, 1.4: 180.0, 1.5: 180.0,  
1.6: 150.0}
```

NOAA curve points for CNS calculations.

5.1.1.3 `yabai.constants.NOAA_CNS_EQUATIONS`

```
yabai.constants.NOAA_CNS_EQUATIONS = {(0.5, 0.6): (-1800.0, 1800.0), (0.6, 0.7):  
(-1500.0, 1620.0), (0.7, 0.8): (-1200.0, 1410.0), (0.8, 0.9): (-900.0, 1170.0), (0.9,  
1.1): (-600.0, 900.0), (1.1, 1.5): (-300.0, 570.0), (1.5, 1.6): (-750.0, 1245.0)}
```

Linealized form of the NOAA CNS curve.

5.1.1.4 `yabai.constants.PW`

```
yabai.constants.PW = 0.0567
```

Water vapour pressure.

Type
float

5.1.2 `yabai.exceptions`

Custom exceptions

Exceptions

<i>InterpolationError</i>	Interpolation error
---------------------------	---------------------

5.1.2.1 `yabai.exceptions.InterpolationError`

exception `yabai.exceptions.InterpolationError`

Interpolation error

5.1.3 `yabai.profile`

Dive profile definitions

Classes

<i>IntegrationPoint</i> (waypoint)	Dive profile integration points.
<i>Parameters</i> ([last_stop_depth, ...])	Dive profile general parameters storage.
<i>Profile</i> (waypoints, tanks[, params])	Dive profile.
<i>Waypoint</i> ([depth, duration, runtime, tank])	Dive profile waypoints.

5.1.3.1 yabai.profile.IntegrationPoint

class yabai.profile.IntegrationPoint(*waypoint*: Waypoint)

Bases: object

Dive profile integration points.

An integration point contains the dive profile information at every calculated point.

Parameters

waypoint – *Waypoint* defining the basic data for calculation.

Attributes

<i>ceiling</i>	Ceiling at the integration point, in meters
<i>p_amb</i>	Absolute ambient pressure at the depth of the integration point, in ATA
<i>waypoint</i>	Waypoint with the depth, duration and runtime of the integration point.
<i>tank_pressure</i>	Pressure of all the tanks at the integration point, in bar.
<i>load_ig</i>	Absolute pressure of inert gas loading in every compartment at the integration point, in ATA.
<i>ceilings</i>	Calculated ceilings for all the compartments at the integration point, in meters.
<i>otu</i>	Oxygen Toxicity Units (OTU) at the integration point.
<i>otu_cum</i>	Accumulated Oxygen Toxicity Units (OTU) at the calculated point.
<i>cns</i>	Central Nervous System oxygen toxicity (CNS) at the integration point, in percentage.
<i>cns_cum</i>	Accumulated Central Nervous System oxygen toxicity (CNS) at the integration point, in percentage.

property ceiling: float

Ceiling at the integration point, in meters

ceilings

Calculated ceilings for all the compartments at the integration point, in meters.

cns

Central Nervous System oxygen toxicity (CNS) at the integration point, in percentage.

cns_cum

Accumulated Central Nervous System oxygen toxicity (CNS) at the integration point, in percentage.

load_ig

Absolute pressure of inert gas loading in every compartment at the integration point, in ATA.

otu

Oxygen Toxicity Units (OTU) at the integration point.

otu_cum

Accumulated Oxygen Toxicity Units (OTU) at the calculated point.

property p_amb: float

Absolute ambient pressure at the depth of the integration point, in ATA

tank_pressure

Pressure of all the tanks at the integration point, in bar.

waypoint

Waypoint with the depth, duration and runtime of the integration point.

5.1.3.2 yabai.profile.Parameters

```
class yabai.profile.Parameters(last_stop_depth: float = 3, stop_depth_incr: float = 3, safety_stop_depth:  
float = 5, safety_stop_duration: float = 3, v_asc: float = 10, v_desc: float =  
20, own_descent_sac: float = 20, own_bottom_sac: float = 20,  
own_ascent_sac: float = 17, buddy_ascent_sac: float = 17, gf_high: float =  
1.0, gf_low: float = 1.0, calc_ascent: bool = True, calc_descent: bool =  
True, deco_stops: bool = True, safety_stop: bool = True, gas_switch_mode:  
str = 'stop', gas_switch_duration: float = 60, dt: timedelta =  
datetime.timedelta(seconds=1))
```

Bases: object

Dive profile general parameters storage.

Attributes

<code>buddy_ascent_sac</code>	Buddy SAC (Surface Air Consumption) during the ascent phase.
<code>calc_ascent</code>	States wether the ascent phase shall be calculated.
<code>calc_descent</code>	States wether the descent phase shall be calculated.
<code>deco_stops</code>	States wether deco stops shall be added during the ascent phase.
<code>dt</code>	Time interval for profile integration
<code>gas_switch_duration</code>	Duration of a gas switch.
<code>gas_switch_mode</code>	{ 'manual', 'stop', 'depth' }
<code>gf_high</code>	Gradient factor high.
<code>gf_low</code>	Gradient factor low.
<code>last_stop_depth</code>	Depth for the last decompression stop.
<code>own_ascent_sac</code>	Diver SAC (Surface Air Consumption) during the ascent phase.
<code>own_bottom_sac</code>	Diver SAC (Surface Air Consumption) during the bottom phase.
<code>own_descent_sac</code>	Diver SAC (Surface Air Consumption) during the descent phase.
<code>safety_stop</code>	States wether a safety stop shall be added at the end of the ascent phase.
<code>safety_stop_depth</code>	Depth for the safety stop.
<code>safety_stop_duration</code>	Duration of the safety stop.
<code>stop_depth_incr</code>	Depth difference between decompression stops.
<code>v_asc</code>	Ascent speed.
<code>v_desc</code>	Descent speed.

buddy_ascent_sac: float = 17

Buddy SAC (Surface Air Consumption) during the ascent phase.

calc_ascent: bool = True

States wether the ascent phase shall be calculated.

calc_descent: bool = True

States whether the descent phase shall be calculated.

deco_stops: bool = True

States whether deco stops shall be added during the ascent phase.

dt: timedelta = datetime.timedelta(seconds=1)

Time interval for profile integration

gas_switch_duration: float = 60

Duration of a gas switch.

gas_switch_mode: str = 'stop'

{ 'manual', 'stop', 'depth' }

Type

Selects the gas switch mode. Allowed values

gf_high: float = 1.0

Gradient factor high.

gf_low: float = 1.0

Gradient factor low.

last_stop_depth: float = 3

Depth for the last decompression stop.

own_ascent_sac: float = 17

Diver SAC (Surface Air Consumption) during the ascent phase.

own_bottom_sac: float = 20

Diver SAC (Surface Air Consumption) during the bottom phase.

own_descent_sac: float = 20

Diver SAC (Surface Air Consumption) during the descent phase.

safety_stop: bool = True

States whether a safety stop shall be added at the end of the ascent phase.

safety_stop_depth: float = 5

Depth for the safety stop.

safety_stop_duration: float = 3

Duration of the safety stop.

stop_depth_incr: float = 3

Depth difference between decompression stops.

v_asc: float = 10

Ascent speed.

v_desc: float = 20

Descent speed.

5.1.3.3 yabai.profile.Profile

```
class yabai.profile.Profile(waypoints: list[Waypoint], tanks: list[Tank], params: Parameters =
    Parameters(last_stop_depth=3, stop_depth_incr=3, safety_stop_depth=5,
    safety_stop_duration=3, v_asc=10, v_desc=20, own_descent_sac=20,
    own_bottom_sac=20, own_ascent_sac=17, buddy_ascent_sac=17,
    gf_high=1.0, gf_low=1.0, calc_ascent=True, calc_descent=True,
    deco_stops=True, safety_stop=True, gas_switch_mode='stop',
    gas_switch_duration=60, dt=datetime.timedelta(seconds=1)))
```

Bases: object

Dive profile.

Parameters

- **waypoints** – List of [Waypoint](#) defining the different profile phases.
- **tanks** – List of [yabai.tanks.Tank](#) used during the dive.
- **params** – [Parameters](#) of the dive.

Methods

plot()	Plots the dive profile along with the ceiling.
plot_ceiling()	Plots the ceiling of dive.
plot_ceilings()	Plots the ceiling of all the compartments.
plot_compartment(gas, compartment)	Plots the loading of the selected gas for the given compartment.
plot_compartments(gas)	Plots the loading of the selected gas for the all the compartments.
plot_integration_points()	Plots the dive profile only using the integration points.
plot_waypoints()	Plots the dive profile using the waypoints.

Attributes

integration_points	List of calculated integration points.
waypoints	List of given and calculated waypoints.

property integration_points: list[[IntegrationPoint](#)]

List of calculated integration points.

plot() → None

Plots the dive profile along with the ceiling.

plot_ceiling() → None

Plots the ceiling of dive.

plot_ceilings() → None

Plots the ceiling of all the compartments.

plot_compartment(gas: str, compartment: int) → None

Plots the loading of the selected gas for the given compartment.

Parameters

- **gas** – Gas to plot. Allowed values: { 'N2' | 'He' }

- **compartment** – Compartment to plot. Allowed values: {1 .. 16}

plot_compartments(*gas*: *str*) → None

Plots the loading of the selected gas for the all the compartments.

Parameters

gas – Gas to plot. Allowed values: {'N2' | 'He'}

plot_integration_points() → None

Plots the dive profile only using the integration points.

plot_waypoints() → None

Plots the dive profile using the waypoints.

property waypoints: **list**[[Waypoint](#)]

List of given and calculated waypoints.

5.1.3.4 yabai.profile.Waypoint

class `yabai.profile.Waypoint`(*depth*: *float* = 0.0, *duration*: *float* | *timedelta* = None, *runtime*: *float* | *timedelta* = None, *tank*: *int* = 0)

Bases: object

Dive profile waypoints.

A waypoint marks a change in the dive profile, i.e. from descent to bottom phases, the start of a deco stop, etc.

Parameters

- **depth** – Depth of the waypoint.
- **duration** – Time interval until the next waypoint.
- **runtime** – Start time counted from the beginning of the dive.
- **tank** – Tank used during until the next waypoint

5.1.4 yabai.tanks

Tank and gas definitions.

Classes

Gas ([o2, he])	Gas mix contained in the scuba tank.
Tank (start_pressure, gas, size)	A scuba tank.

5.1.4.1 yabai.tanks.Gas

class `yabai.tanks.Gas`(*o2*: *int* = 21, *he*: *int* = 0)

Bases: object

Gas mix contained in the scuba tank.

Parameters

- **o2** – Oxygen content in the gas mixture, expressed in percentage, defaults to 21.
- **he** – Helium content in the gas mixture, expressed in percentage, defaults to 0.

Methods

<code>mod([pp_o2])</code>	Calculates the Maximum Operating Depth (MOD) of the gas mix at a given oxygen partial
<code>ppHe(depth)</code>	Calculates the helium partial pressure of the gas mix at a given depth.
<code>ppN2(depth)</code>	Calculates the nitrogen partial pressure of the gas mix at a given depth.
<code>ppO2(depth)</code>	Calculates the oxygen partial pressure of the gas mix at a given depth.

Attributes

<code>He</code>	Helium content of the gas mix (in percentage).
<code>N2</code>	Nitrogen content of the gas mix (in percentage).
<code>O2</code>	Oxygen content of the gas mix (in percentage).
<code>fHe</code>	Helium fraction of the gas mix (decimal).
<code>fN2</code>	Nitrogen fraction of the gas mix (decimal).
<code>fO2</code>	Oxygen fraction of the gas mix (decimal).

property He: int

Helium content of the gas mix (in percentage).

property N2: int

Nitrogen content of the gas mix (in percentage).

property O2: int

Oxygen content of the gas mix (in percentage).

property fHe: float

Helium fraction of the gas mix (decimal).

property fN2: float

Nitrogen fraction of the gas mix (decimal).

property fO2: float

Oxygen fraction of the gas mix (decimal).

mod(pp_o2=1.4) → float

Calculates the Maximum Operating Depth (MOD) of the gas mix at a given oxygen partial pressure.

Parameters

pp_o2 – Oxygen partial pressure to calculate MOD.

Returns

The maximum operating depth of the gas mix at the given oxygen partial pressure.

ppHe(depth: float) → float

Calculates the helium partial pressure of the gas mix at a given depth.

Parameters

depth – Depth to calculate helium partial pressure of the gas mix (in meters).

Returns

The helium partial pressure of the gas mix at the given depth.

ppN2(*depth: float*) → float

Calculates the nitrogen partial pressure of the gas mix at a given depth.

Parameters

depth – Depth to calculate nitrogen partial pressure of the gas mix (in meters).

Returns

The nitrogen partial pressure of the gas mix at the given depth.

ppO2(*depth: float*) → float

Calculates the oxygen partial pressure of the gas mix at a given depth.

Parameters

depth – Depth to calculate oxygen partial pressure of the gas mix (in meters).

Returns

The oxygen partial pressure of the gas mix at the given depth.

5.1.4.2 yabai.tanks.Tank

class yabai.tanks.**Tank**(*start_pressure: int = 200, gas: ~yabai.tanks.Gas = <Gas Mixture: O2: 21 N2: 79 He: 0>, size: int = 15*)

Bases: object

A scuba tank.

Parameters

- **start_pressure** – Tank start pressure.
- **gas** – Gas mix in the tank.
- **size** – Size of the tank, in liters.

Attributes

<i>gas</i>	Gas mix in the tank.
<i>size</i>	Size of the tank, in liters.

property gas: *Gas*

Gas mix in the tank.

property size: *int*

Size of the tank, in liters.

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